

## Project Initialization and Planning Phase

|               |                                 |
|---------------|---------------------------------|
| Date          | 7 July 2026                     |
| Team ID       | SWTID-2026-2018                 |
| Project Title | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Project Proposal (Proposed Solution) Report

#### Project Proposal (Proposed Solution)

The proposed solution aims to transform the credit card eligibility assessment process using Machine Learning to improve efficiency, consistency, and decision-making. Traditional credit card application screening often involves manual verification, making the process time-consuming and prone to human error. The proposed system automates the preliminary eligibility assessment by analyzing applicant demographic, financial, and credit-related information using machine learning algorithms. The application provides instant eligibility predictions, confidence scores, and personalized recommendations through a user-friendly web interface, helping both applicants and financial institutions make informed decisions.

| Section   | Details                                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective | The primary objective is to develop an intelligent web-based system that predicts credit card eligibility using machine learning techniques, enabling faster, more accurate, and consistent preliminary assessments.        |
| Scope     | The project covers data preprocessing, feature engineering, machine learning model training, model evaluation, and deployment of a Flask-based web application that provides real-time credit card eligibility predictions. |

#### Problem Statement

| Section     | Details                                                                                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Banks and financial institutions receive a large number of credit card applications every day. Manual screening is time-consuming, inconsistent, and may result in delays or incorrect decisions. Applicants also lack a reliable way to estimate their eligibility before submitting an application. |
| Impact      | Automating the eligibility assessment process reduces manual effort, improves consistency in decision-making, speeds up application screening, enhances customer experience, and helps financial institutions identify high-risk applicants early.                                                    |

## Proposed Solution

| Section      | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach     | Build a machine learning-based web application that analyzes applicant information, preprocesses the data, and predicts credit card eligibility using a trained XGBoost classification model. The system provides real-time predictions through a Flask web interface.                                                                                                                                                                                                                                                                 |
| Key Features | <ul style="list-style-type: none"><li>• Machine learning-based credit card eligibility prediction.</li><li>• Automated data preprocessing and feature engineering.</li><li>• Comparison of multiple classification algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost).</li><li>• Real-time eligibility prediction with confidence score and risk level.</li><li>• Personalized recommendations for applicants.</li><li>• Responsive Flask web application.</li><li>• Cloud deployment using Render.</li></ul> |

## Resource Requirements

| Resource Type | Description             | Specification / Allocation                                                                                                |
|---------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Hardware      | Computing Resources     | Intel Core i3/i5/i7 Processor or equivalent                                                                               |
|               | Memory                  | Minimum 8 GB RAM                                                                                                          |
|               | Storage                 | Minimum 20 GB Free Disk Space                                                                                             |
| Software      | Framework               | Flask                                                                                                                     |
|               | Programming Language    | Python 3.10+                                                                                                              |
|               | Libraries               | Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib, OpenPyXL                                               |
|               | Development Environment | Visual Studio Code / PyCharm                                                                                              |
|               | Version Control         | Git & GitHub                                                                                                              |
|               | Deployment Platform     | Render with Gunicorn                                                                                                      |
| Data          | Dataset                 | CreditCardApproval_Selected_Features.csv                                                                                  |
|               | Source                  | Kaggle (Credit Card Approval Dataset)                                                                                     |
|               | Format                  | CSV                                                                                                                       |
|               | Records                 | Approximately 307,000+ applicant records                                                                                  |
|               | Features                | Selected demographic, financial, employment, housing, and credit inquiry attributes used for machine learning prediction. |